

DAY-9

 jupyter

practice prblm Last Checkpoint: 2 days ago



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 +      Code

JupyterLab  Python 3 (ipykernel)

```
[89]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_excel(
    "Supermarket-Sales-Sample-Data (1).xlsx",
    header=7
)
print(df.head())
```

Unnamed: 0	Order No	Order Date	Customer Name	Ship Date	
0	NaN	1001 2024-01-01	John Smith	2024-01-03	
1	NaN	1002 2024-01-01	Jane Doe	2024-01-04	
2	NaN	1003 2024-01-02	Michael Johnson	2024-01-07	
3	NaN	1004 2024-01-02	Emily Brown	2024-01-03	
4	NaN	1005 2024-01-03	David Wilson	2024-01-08	

	Retail Price (USD)	Order Quantity	Tax (USD)	Total (USD)
0	49.99	2	9.998	109.978
1	29.99	1	2.999	32.989
2	99.99	3	29.997	329.967
3	19.99	4	7.996	87.956
4	149.99	1	14.999	164.989

```
[90]: print(df.isnull())
print(df.head(5))
```

Unnamed: 0	Order No	Order Date	Customer Name	Ship Date	
0	True	False	False	False	False
1	True	False	False	False	False
2	True	False	False	False	False
3	True	False	False	False	False
4	True	False	False	False	False
..	...	...	...	...	...
65	True	False	False	False	False
66	True	False	False	False	False
67	True	False	False	False	False
68	True	False	False	False	False
69	True	False	False	False	False

0	True	False	False	False	False
1	True	False	False	False	False
2	True	False	False	False	False
3	True	False	False	False	False
4	True	False	False	False	False
..	...	...	...	...	...
65	True	False	False	False	False
66	True	False	False	False	False
67	True	False	False	False	False
68	True	False	False	False	False
69	True	False	False	False	False

	Retail Price (USD)	Order Quantity	Tax (USD)	Total (USD)
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False
..	...	...	...	...
65	False	False	False	False
66	False	False	False	False
67	False	False	False	False
68	False	False	False	False
69	False	False	False	False

```
[70 rows x 9 columns]
```

Unnamed: 0	Order No	Order Date	Customer Name	Ship Date	
0	NaN	1001 2024-01-01	John Smith	2024-01-03	
1	NaN	1002 2024-01-01	Jane Doe	2024-01-04	
2	NaN	1003 2024-01-02	Michael Johnson	2024-01-07	
3	NaN	1004 2024-01-02	Emily Brown	2024-01-03	
4	NaN	1005 2024-01-03	David Wilson	2024-01-08	

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2	99.99	3	29.997	329.967
3	19.99	4	7.996	87.956
4	149.99	1	14.999	164.989

```
[91]: df.isnull().sum()
```

```
[91]: Unnamed: 0      70
      Order No      0
      Order Date    0
      Customer Name  0
      Ship Date     0
      Retail Price (USD)  0
      Order Quantity 0
      Tax (USD)      0
      Total (USD)    0
      dtype: int64
```

```
[92]: print(df.columns)
```

```
Index(['Unnamed: 0', 'Order No', 'Order Date', 'Customer Name', 'Ship Date',
      'Retail Price (USD)', 'Order Quantity', 'Tax (USD)', 'Total (USD)'],
      dtype='str')
```

```
[93]: df.rename(columns={'Unnamed: 0': 'Sno'}, inplace=True)
      print(df.columns)
```

```
Index(['Sno', 'Order No', 'Order Date', 'Customer Name', 'Ship Date',
      'Retail Price (USD)', 'Order Quantity', 'Tax (USD)', 'Total (USD)'],
      dtype='str')
```

```
[94]: df['Sno'] = range(1, len(df) + 1)
      print(df.head())
```

	Sno	Order No	Order Date	Customer Name	Ship Date	Retail Price (USD)	\
0	1	1001	2024-01-01	John Smith	2024-01-03	49.99	
1	2	1002	2024-01-01	Jane Doe	2024-01-04	29.99	
2	3	1003	2024-01-02	Michael Johnson	2024-01-07	99.99	
3	4	1004	2024-01-02	Emily Brown	2024-01-03	19.99	
4	5	1005	2024-01-03	David Wilson	2024-01-08	149.99	

2	3	1003	2024-01-02	Michael Johnson	2024-01-07	99.99
3	4	1004	2024-01-02	Emily Brown	2024-01-03	19.99
4	5	1005	2024-01-03	David Wilson	2024-01-08	149.99

	Order Quantity	Tax (USD)	Total (USD)
0	2	9.998	109.978
1	1	2.999	32.989
2	3	29.997	329.967
3	4	7.996	87.956
4	1	14.999	164.989

```
[95]: duplicates = df[df.duplicated()]
      print(duplicates)
```

```
Empty DataFrame
Columns: [Sno, Order No, Order Date, Customer Name, Ship Date, Retail Price (USD), Order Quantity, Tax (USD), Total (USD)]
Index: []
```

```
[96]: df.info()
```

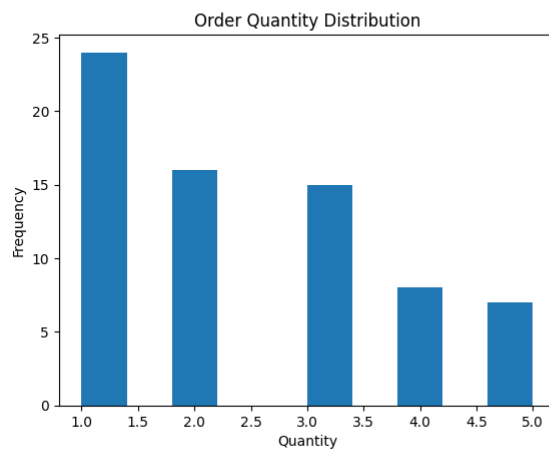
```
<class 'pandas.DataFrame'>
RangeIndex: 70 entries, 0 to 69
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   Sno                 70 non-null    int64
 1   Order No            70 non-null    int64
 2   Order Date          70 non-null    datetime64[us]
 3   Customer Name       70 non-null    str
 4   Ship Date           70 non-null    datetime64[us]
 5   Retail Price (USD)  70 non-null    float64
 6   Order Quantity      70 non-null    int64
 7   Tax (USD)           70 non-null    float64
 8   Total (USD)         70 non-null    float64
dtypes: datetime64[us](2), float64(3), int64(3), str(1)
memory usage: 5.1 KB
```

```
[97]: import matplotlib.pyplot as plt
```

```
plt.hist(df['Order Quantity'])
```

```
[97]: import matplotlib.pyplot as plt
```

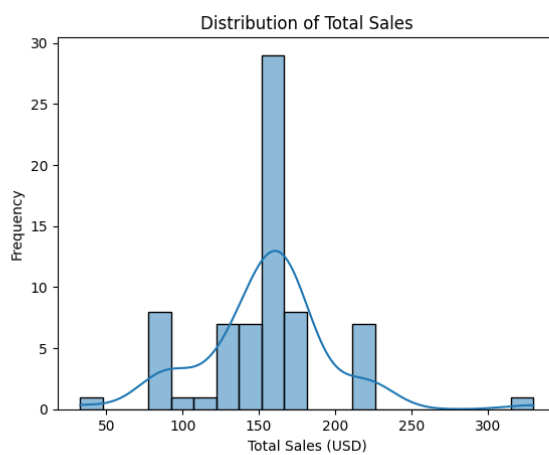
```
plt.hist(df['Order Quantity'])  
plt.title('Order Quantity Distribution')  
plt.xlabel('Quantity')  
plt.ylabel('Frequency')  
plt.show()
```



```
[98]: sns.histplot(df['Total (USD)'], bins=20, kde=True)
```

```
[98]: sns.histplot(df['Total (USD)'], bins=20, kde=True)
```

```
plt.title("Distribution of Total Sales")  
plt.xlabel("Total Sales (USD)")  
plt.ylabel("Frequency")  
plt.show()
```



```
[99]:
```

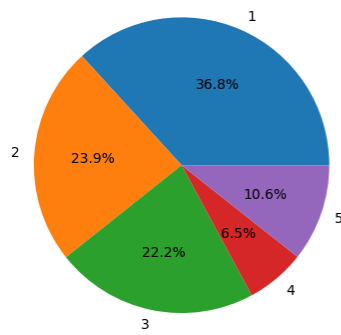
Total Sales (USD)

```
[99]: labels = sales.index
      values = sales.values

      plt.pie(
          values,
          labels=labels,
          autopct='%1.1f%%'
      )

      plt.title("Sales Distribution by Customer")
      plt.show()
```

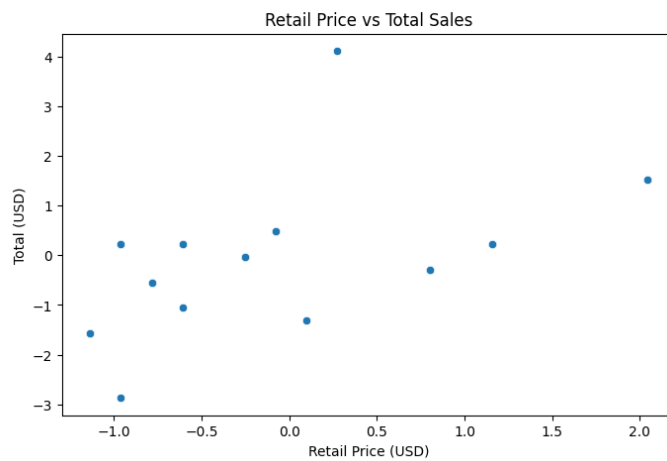
Sales Distribution by Customer



```
[106]: plt.figure(figsize=(8,5))

      sns.scatterplot(
          x='Retail Price (USD)',
          y='Total (USD)',
          data=df
      )

      plt.title("Retail Price vs Total Sales")
      plt.show()
```





```
[109]: from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
cols = ['Retail Price (USD)', 'Order Quantity', 'Total (USD)', 'Tax (USD)']
df[cols] = scaler.fit_transform(df[cols])
print(df.head())
```

	Sno	Order No	Order Date	Customer Name	Ship Date	Retail Price (USD)	\
0	1	1001	2024-01-01	John Smith	2024-01-03	-0.607902	
1	2	1002	2024-01-01	Jane Doe	2024-01-04	-0.961040	
2	3	1003	2024-01-02	Michael Johnson	2024-01-07	0.274943	
3	4	1004	2024-01-02	Emily Brown	2024-01-03	-1.137609	
4	5	1005	2024-01-03	David Wilson	2024-01-08	1.157789	

	Order Quantity	Tax (USD)	Total (USD)
0	-0.302002	-1.060285	-1.060285
1	-1.057007	-2.870446	-2.870446
2	0.453003	4.112083	4.112083
3	1.208008	-1.578065	-1.578065
4	-1.057007	0.233130	0.233130

[]:

